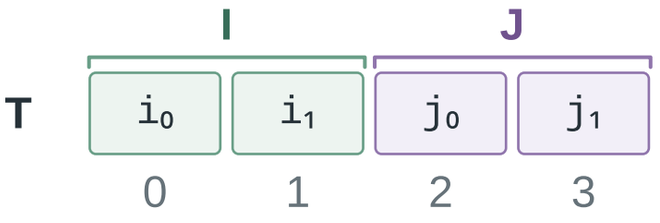
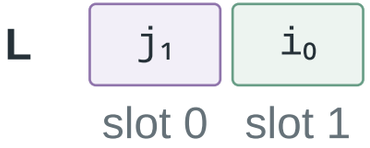
`dot(dot(X, Y), Z) → dot(X, dot(Y, Z))`

T = dot(X, Y), O = dot(T, Z). Inner dot contracts K; outer dot contracts L.

1. Select axes



`c_L = [3, 0]`



Tensor order is unchanged

→

2. Partition by origin

`I = [i₀, i₁]`

`I_sel [i₀]` `I_rem [i₁]`

`v_I = [0], u_I = [1]`

`J = [j₀, j₁]`

`J_sel [j₁]` `J_rem [j₀]`

`v_J = [1], u_J = [0]`

Views retain axis positions

→

3. Preserve pairings

Physical Z axes



`c_R = [2, 0]`

Slot	From T		From Z	Z pos
0	j_1	↔	z_1	2
1	i_0	↔	z_0	0

Selected subgroup mappings

Positions are zero-based

Selected group	Producer positions	Slots in L	Partner positions in Z
`I_sel`	`v_I`	`u_I`	`gather(c_R, u_I)`
`J_sel`	`v_J`	`u_J`	`gather(c_R, u_J)`

U = dot(Y, Z) contracts J_sel; final dot contracts I_sel + K.

U retains K and I_sel's partners. Output: I_rem ++ J_rem ++ F (F = [f]).